

EXAMPLES

STEP 1: CREATE PHYSICIAN TABLES

```
[mysql> USE student_db
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> CREATE TABLE physician (
  ->     employeeid INT PRIMARY KEY,
  ->     name VARCHAR(100),
  ->     position VARCHAR(100),
  ->     ssn VARCHAR(20)
[  -> );
Query OK, 0 rows affected (0.018 sec)
```

STEP 2: INSERT DATA INTO PHYSICIAN

```
mysql> INSERT INTO physician VALUES
  -> (1,'John Dorian','Staff Internist','111111111'),
  -> (2,'Elliot Reid','Attending Physician','222222222'),
  -> (3,'Christopher Turk','Surgical Attending Physician','333333333'),
  -> (4,'Percival Cox','Senior Attending Physician','444444444'),
  -> (5,'Bob Kelso','Head Chief of Medicine','555555555'),
  -> (6,'Todd Quinlan','Surgical Attending Physician','666666666'),
  -> (7,'John Wen','Surgical Attending Physician','777777777'),
  -> (8,'Keith Dudemeister','MD Resident','888888888'),
  -> (9,'Molly Clock','Attending Psychiatrist','999999999');
Query OK, 9 rows affected (0.003 sec)
Records: 9  Duplicates: 0  Warnings: 0

[mysql> SELECT * FROM physician;
+-----+-----+-----+-----+
| employeeid | name           | position                | ssn      |
+-----+-----+-----+-----+
| 1          | John Dorian    | Staff Internist         | 111111111 |
| 2          | Elliot Reid    | Attending Physician     | 222222222 |
| 3          | Christopher Turk | Surgical Attending Physician | 333333333 |
| 4          | Percival Cox   | Senior Attending Physician | 444444444 |
| 5          | Bob Kelso      | Head Chief of Medicine  | 555555555 |
| 6          | Todd Quinlan   | Surgical Attending Physician | 666666666 |
| 7          | John Wen       | Surgical Attending Physician | 777777777 |
| 8          | Keith Dudemeister | MD Resident            | 888888888 |
| 9          | Molly Clock    | Attending Psychiatrist   | 999999999 |
+-----+-----+-----+-----+
9 rows in set (0.000 sec)
```

STEP 3: CREATE DEPARTMENT TABLE

```
mysql> CREATE TABLE department (
  ->     departmentid INT PRIMARY KEY,
  ->     name VARCHAR(100),
  ->     head INT
[  -> );
Query OK, 0 rows affected (0.008 sec)
```

STEP 4: INSERT DATA INTO DEPARTMENT

```
mysql> INSERT INTO department VALUES
    -> (1,'General Medicine',4),
    -> (2,'Surgery',7),
    -> (3,'Psychiatry',9);
Query OK, 3 rows affected (0.003 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
[mysql> SELECT * FROM department;

+-----+-----+-----+
| departmentid | name           | head |
+-----+-----+-----+
|          1 | General Medicine |    4 |
|          2 | Surgery         |    7 |
|          3 | Psychiatry      |    9 |
+-----+-----+-----+
3 rows in set (0.000 sec)
```

QUESTION 1:- INNER JOIN

```
mysql> SELECT d.name AS Department,
    ->          p.name AS Head_Physician
    -> FROM department d
    -> JOIN physician p
    -> ON d.head = p.employeeid;

+-----+-----+
| Department | Head_Physician |
+-----+-----+
| General Medicine | Percival Cox |
| Surgery         | John Wen      |
| Psychiatry      | Molly Clock   |
+-----+-----+
3 rows in set (0.001 sec)
```

QUESTION 2:- WHERE CLAUSE

```
mysql> SELECT *
    -> FROM physician
    -> WHERE position = 'Surgical Attending Physician';

+-----+-----+-----+-----+
| employeeid | name           | position                               | ssn       |
+-----+-----+-----+-----+
|          3 | Christopher Turk | Surgical Attending Physician | 333333333 |
|          6 | Todd Quinlan    | Surgical Attending Physician | 666666666 |
|          7 | John Wen        | Surgical Attending Physician | 777777777 |
+-----+-----+-----+-----+
3 rows in set (0.001 sec)
```

QUESTION 3:- WHERE + LIKE

```
mysql> SELECT *
-> FROM physician
-> WHERE name LIKE 'John%';
```

employeeid	name	position	ssn
1	John Dorian	Staff Internist	111111111
7	John Wen	Surgical Attending Physician	777777777

2 rows in set (0.002 sec)

QUESTION 4:- COUNT()

```
mysql> SELECT COUNT(*) AS Total_Physicians
-> FROM physician;
```

Total_Physicians
9

1 row in set (0.001 sec)

QUESTION 5:- COUNT DISTINCT

```
mysql> SELECT COUNT(DISTINCT position)
-> FROM physician;
```

COUNT(DISTINCT position)
7

1 row in set (0.001 sec)

QUESTION 6:- GROUP BY

```
mysql> SELECT position,
-> COUNT(*) AS Total_Employees
-> FROM physician
-> GROUP BY position;
```

position	Total_Employees
Staff Internist	1
Attending Physician	1
Surgical Attending Physician	3
Senior Attending Physician	1
Head Chief of Medicine	1
MD Resident	1
Attending Psychiatrist	1

7 rows in set (0.001 sec)

QUESTION 7:- GROUP BY + HAVING

```
mysql> SELECT position,  
->      COUNT(*) AS Total_Employees  
-> FROM physician  
-> GROUP BY position  
-> HAVING COUNT(*) > 1;
```

position	Total_Employees
Surgical Attending Physician	3

1 row in set (0.001 sec)

QUESTION 8:- ORDER BY

```
mysql> SELECT *  
-> FROM physician  
-> ORDER BY name ASC;
```

employeeid	name	position	ssn
5	Bob Kelso	Head Chief of Medicine	555555555
3	Christopher Turk	Surgical Attending Physician	333333333
2	Elliot Reid	Attending Physician	222222222
1	John Dorian	Staff Internist	111111111
7	John Wen	Surgical Attending Physician	777777777
8	Keith Dudemeister	MD Resident	888888888
9	Molly Clock	Attending Psychiatrist	999999999
4	Percival Cox	Senior Attending Physician	444444444
6	Todd Quinlan	Surgical Attending Physician	666666666

9 rows in set (0.002 sec)

QUESTION 9:- ORDER BY DESC

```
mysql> SELECT *  
-> FROM physician  
-> ORDER BY employeeid DESC;
```

employeeid	name	position	ssn
9	Molly Clock	Attending Psychiatrist	999999999
8	Keith Dudemeister	MD Resident	888888888
7	John Wen	Surgical Attending Physician	777777777
6	Todd Quinlan	Surgical Attending Physician	666666666
5	Bob Kelso	Head Chief of Medicine	555555555
4	Percival Cox	Senior Attending Physician	444444444
3	Christopher Turk	Surgical Attending Physician	333333333
2	Elliot Reid	Attending Physician	222222222
1	John Dorian	Staff Internist	111111111

9 rows in set (0.000 sec)

QUESTION 10:- WHERE + IN

```
mysql> SELECT *
  -> FROM physician
  -> WHERE employeeid IN
  -> (
  ->     SELECT head
  ->     FROM department
  -> );
```

employeeid	name	position	ssn
4	Percival Cox	Senior Attending Physician	444444444
7	John Wen	Surgical Attending Physician	777777777
9	Molly Clock	Attending Psychiatrist	999999999

3 rows in set (0.002 sec)

QUESTION 11:- SUBQUERY

```
mysql> SELECT *
  -> FROM physician
  -> WHERE employeeid NOT IN
  -> (
  ->     SELECT head
  ->     FROM department
  -> );
```

employeeid	name	position	ssn
1	John Dorian	Staff Internist	111111111
2	Elliot Reid	Attending Physician	222222222
3	Christopher Turk	Surgical Attending Physician	333333333
5	Bob Kelso	Head Chief of Medicine	555555555
6	Todd Quinlan	Surgical Attending Physician	666666666
8	Keith Dudemeister	MD Resident	888888888

6 rows in set (0.001 sec)

QUESTION 12:- GROUP BY + ORDER BY

```
mysql> SELECT position,
  ->     COUNT(*) AS Total_Employees
  -> FROM physician
  -> GROUP BY position
  -> ORDER BY Total_Employees DESC;
```

position	Total_Employees
Surgical Attending Physician	3
Staff Internist	1
Attending Physician	1
Senior Attending Physician	1
Head Chief of Medicine	1
MD Resident	1
Attending Psychiatrist	1

7 rows in set (0.001 sec)

QUESTION 13:- WHERE + LIKE

```
mysql> SELECT *
  -> FROM physician
  -> WHERE position LIKE '%Attending%';
```

employeeid	name	position	ssn
2	Elliot Reid	Attending Physician	222222222
3	Christopher Turk	Surgical Attending Physician	333333333
4	Percival Cox	Senior Attending Physician	444444444
6	Todd Quinlan	Surgical Attending Physician	666666666
7	John Wen	Surgical Attending Physician	777777777
9	Molly Clock	Attending Psychiatrist	999999999

6 rows in set (0.001 sec)

QUESTION 14:- JOIN + WHERE

```
mysql> SELECT p.name
  -> FROM physician p
  -> JOIN department d
  -> ON p.employeeid = d.head
  -> WHERE d.name = 'Surgery';
```

name
John Wen

1 row in set (0.001 sec)

QUESTION 15:- JOIN + ORDER BY

```
mysql> SELECT d.name,
  -> p.name
  -> FROM department d
  -> JOIN physician p
  -> ON d.head = p.employeeid
  -> ORDER BY d.name;
```

name	name
General Medicine	Percival Cox
Psychiatry	Molly Clock
Surgery	John Wen

3 rows in set (0.001 sec)

QUESTION 16:- WHERE + NOT LIKE

```
mysql> SELECT *
-> FROM physician
-> WHERE position NOT LIKE '%Surgical%';
```

employeeid	name	position	ssn
1	John Dorian	Staff Internist	111111111
2	Elliot Reid	Attending Physician	222222222
4	Percival Cox	Senior Attending Physician	444444444
5	Bob Kelso	Head Chief of Medicine	555555555
8	Keith Dudemeister	MD Resident	888888888
9	Molly Clock	Attending Psychiatrist	999999999

6 rows in set (0.000 sec)

QUESTION 17:- COUNT + HAVING

```
mysql> SELECT position,
-> COUNT(*) AS Total
-> FROM physician
-> GROUP BY position
-> HAVING COUNT(*) >= 2;
```

position	Total
Surgical Attending Physician	3

1 row in set (0.001 sec)

QUESTION 18:- EXISTS

```
mysql> SELECT *
-> FROM department d
-> WHERE EXISTS
-> (
-> SELECT 1
-> FROM physician p
-> WHERE p.employeeid = d.head
-> );
```

departmentid	name	head
1	General Medicine	4
2	Surgery	7
3	Psychiatry	9

3 rows in set (0.001 sec)

QUESTION 19:- MAX()

```
+-----+-----+-----+
| departmentid | name          | head |
+-----+-----+-----+
|          1 | General Medicine | 4    |
|          2 | Surgery         | 7    |
|          3 | Psychiatry      | 9    |
+-----+-----+-----+
3 rows in set (0.001 sec)
```

mysql> SELECT MAX(employeeid)
-> FROM physician;

```
+-----+
| MAX(employeeid) |
+-----+
|          9      |
+-----+
1 row in set (0.000 sec)
```

QUESTION 20:- MIN()

```
mysql> SELECT MIN(employeeid)  
-> FROM physician;
```

```
+-----+
| MIN(employeeid) |
+-----+
|          1      |
+-----+
1 row in set (0.000 sec)
```